

1.

What does PowerShell primarily provide?

- a) A web browser
- b) A command-line shell and scripting language
- c) A database management system
- d) A text editor

Answer: b) A command-line shell and scripting language

Explanation: PowerShell is a shell and scripting tool for task automation.

2.

Which symbol denotes a variable in PowerShell?

- a) %
- b) \$
- c) #
- d) @

Answer: b) \$

Explanation: Variables are referenced with \$, e.g., \$myVar.

3.

Which cmdlet is used to list all available cmdlets?

- a) Get-Command
- b) List-Cmdlets
- c) Show-Help
- d) Find-Command

Answer: a) Get-Command

Explanation: Get-Command shows all cmdlets, functions, and aliases.

4.

What does the pipeline operator (|) do in PowerShell?

- a) Comments out code

- b) Runs commands sequentially
- c) Sends output of one cmdlet as input to another
- d) Declares a function

Answer: c) Sends output of one cmdlet as input to another

Explanation: Pipeline chains commands to process objects.

5.

How do you get help for a cmdlet named Get-Process?

- a) Help Get-Process
- b) Get-Help Get-Process
- c) Show-Help Get-Process
- d) HelpCmd Get-Process

Answer: b) Get-Help Get-Process

Explanation: Get-Help provides documentation on cmdlets.

6.

Which of these is a correct way to assign a string variable in PowerShell?

- a) \$name = "John"
- b) name = "John"
- c) %name = 'John'
- d) #name = 'John'

Answer: a) \$name = "John"

Explanation: Variables start with \$ and strings can use double quotes.

7.

Which cmdlet lists all running processes?

- a) Get-Services
- b) Get-Process
- c) Show-Process
- d) List-Processes

Answer: b) Get-Process

Explanation: Get-Process lists current processes on the system.

8.

What is the default execution policy in PowerShell?

- a) Unrestricted
- b) RemoteSigned
- c) Restricted
- d) AllSigned

Answer: c) Restricted

Explanation: The default blocks script execution for safety.

9.

Which cmdlet changes the PowerShell execution policy?

- a) Set-ExecutionPolicy
- b) Change-Policy
- c) New-ExecutionPolicy
- d) Execution-Set

Answer: a) Set-ExecutionPolicy

Explanation: Used to modify script execution restrictions.

10.

Which of the following is a valid PowerShell cmdlet format?

- a) Get-Item
- b) Item-Get
- c) Fetch-Item
- d) Cmd-Item

Answer: a) Get-Item

Explanation: Cmdlets follow Verb-Noun naming convention.

Medium (20)

11.

What type of objects does PowerShell pipeline transmit?

- a) Text strings
- b) Byte arrays
- c) .NET objects
- d) XML only

Answer: c) .NET objects

Explanation: PowerShell pipelines objects, enabling rich data manipulation.

12.

Which cmdlet would you use to import a PowerShell module named "ActiveDirectory"?

- a) Import-Module ActiveDirectory
- b) Add-Module ActiveDirectory
- c) Load-Module ActiveDirectory
- d) Use-Module ActiveDirectory

Answer: a) Import-Module ActiveDirectory

Explanation: Import-Module loads modules into the session.

13.

How do you start an interactive remote session to a computer named Server01?

- a) Enter-PSSession -ComputerName Server01
- b) Invoke-Command Server01
- c) Start-RemoteSession Server01

d) Connect-Remote Server01

Answer: a) Enter-PSSession -ComputerName Server01

Explanation: Enter-PSSession opens an interactive remote shell.

14.

Which cmdlet lists all running services on a system?

a) Get-Process

b) Get-Service

c) Show-Services

d) List-Service

Answer: b) Get-Service

Explanation: Get-Service lists Windows services and their status.

15.

How would you filter all services that are running using pipeline?

a) Get-Service | Where-Object {\$_.Status -eq 'Running'}

b) Get-Service | Filter-Object Running

c) Get-Service -Status Running

d) Get-Service | Select-Running

Answer: a) Get-Service | Where-Object {\$_.Status -eq 'Running'}

Explanation: Where-Object filters objects based on a condition.

16.

Which cmdlet creates a new directory?

a) New-Directory

b) New-Item -ItemType Directory

c) Create-Folder

d) New-Folder

Answer: b) New-Item -ItemType Directory

Explanation: New-Item creates files or folders based on the item type.

17.

What is the output of this command?

Get-Process | Sort-Object CPU -Descending | Select-Object -First 1

- a) Process using the most CPU
- b) Process using the least CPU
- c) All processes sorted ascending by CPU
- d) First process alphabetically

Answer: a) Process using the most CPU

Explanation: Sorts processes by CPU descending, selects the top one.

18.

What is the cmdlet to stop a running process named "notepad"?

- a) Stop-Process -Name notepad
- b) Kill-Process notepad
- c) Terminate-Process notepad
- d) End-Process notepad

Answer: a) Stop-Process -Name notepad

Explanation: Stop-Process terminates a running process.

19.

Which statement is true about Get-Help?

- a) Shows help only for cmdlets
- b) Can update help content using Update-Help
- c) Is deprecated in PowerShell 7
- d) Only works offline

Answer: b) Can update help content using Update-Help

Explanation: You can update help files for cmdlets using Update-Help.

20.

How do you run a PowerShell script named script.ps1 in the current directory?

- a) .\script.ps1
- b) run script.ps1
- c) start script.ps1
- d) execute script.ps1

Answer: a) .\script.ps1

Explanation: .\ tells PowerShell to run the script in the current directory.

21.

Which parameter forces an error to stop execution in PowerShell?

- a) -ErrorAction Stop
- b) -ForceError
- c) -StopOnError
- d) -BreakOnError

Answer: a) -ErrorAction Stop

Explanation: It instructs cmdlets to stop on errors instead of continuing.

22.

What type of data structure is this?

\$hash = @{ Name = "John"; Age = 30 }

- a) Array
- b) Hashtable
- c) List
- d) Dictionary

Answer: b) Hashtable

Explanation: The @{ syntax creates a hashtable in PowerShell.

23.

Which cmdlet is used to schedule tasks in PowerShell?

- a) Register-ScheduledJob

- b) New-Task
- c) Set-CronJob
- d) Schedule-Task

Answer: a) Register-ScheduledJob

Explanation: Used to create scheduled background jobs.

24.

What does Select-Object do?

- a) Filters objects based on property values
- b) Selects specific properties of an object
- c) Sorts objects
- d) Removes duplicates

Answer: b) Selects specific properties of an object

Explanation: Select-Object lets you pick properties or a subset of objects.

25.

Which command shows the current execution policy?

- a) Get-ExecutionPolicy
- b) Show-ExecutionPolicy
- c) Execution-Policy-Get
- d) Display-ExecutionPolicy

Answer: a) Get-ExecutionPolicy

Explanation: Displays the current script execution policy.

26.

What does Try/Catch do in PowerShell?

- a) Defines a loop
- b) Defines error handling blocks
- c) Creates a function
- d) Filters objects

Answer: b) Defines error handling blocks

Explanation: Used to handle exceptions and errors.

27.

Which of the following commands creates an empty text file named log.txt?

a) New-Item -Path . -Name log.txt -ItemType File

b) Create-File log.txt

c) Touch log.txt

d) Set-Content log.txt ""

Answer: a) New-Item -Path . -Name log.txt -ItemType File

Explanation: New-Item creates an empty file at specified path.

28.

What cmdlet is used to get event logs on Windows?

a) Get-EventLog

b) Show-Events

c) Get-Logs

d) Show-EventLog

Answer: a) Get-EventLog

Explanation: Used to retrieve system event logs.

29.

Which cmdlet outputs the current date and time?

a) Get-Date

b) Show-Time

c) Date-Now

d) Get-Time

Answer: a) Get-Date

Explanation: Returns the current system date/time.

30.

How do you comment a single line in PowerShell?

- a) # This is a comment
- b) // This is a comment
- c) <!-- This is a comment -->
- d) /* This is a comment */

Answer: a) # This is a comment

Explanation: # denotes a comment line.

Hard (10)

31.

What does this script do?

powershell

Copy code

```
$services = Get-Service | Where-Object {$_.Status -eq "Stopped"} foreach ($service in $services) { Start-Service $service.Name }
```

- a) Stops all running services
- b) Starts all stopped services
- c) Restarts all services
- d) Lists all services

Answer: b) Starts all stopped services

Explanation: Filters stopped services, loops through and starts them.

32.

Which cmdlet allows you to export objects to a CSV file?

- a) Export-Csv
- b) Save-Csv
- c) Write-Csv
- d) Out-Csv

Answer: a) Export-Csv

Explanation: Serializes objects into CSV format.

33.

In PowerShell remoting, which protocol is primarily used?

- a) SSH
- b) WS-Management (WinRM)
- c) FTP
- d) Telnet

Answer: b) WS-Management (WinRM)

Explanation: PowerShell remoting uses WS-Management via WinRM.

34.

What is the difference between -eq and -like operators?

- a) -eq is for exact match, -like uses wildcard patterns
- b) Both are identical

c) -like is for numbers, -eq is for strings

d) -eq uses regex, -like uses wildcards

Answer: a) -eq is for exact match, -like uses wildcard patterns

Explanation: -like supports * and ? wildcards.

35.

Which of these commands will safely stop a service and handle errors?

powershell

Copy code

```
try { Stop-Service -Name Spooler -ErrorAction Stop } catch { Write-Host "Failed to stop service: $_" }
```

a) Stops Spooler and writes error if failed

b) Always stops Spooler

c) Ignores errors silently

d) Restarts Spooler

Answer: a) Stops Spooler and writes error if failed

Explanation: Uses try/catch with error action stop.

36.

What is the purpose of Start-Job in PowerShell?

a) Starts a background job asynchronously

b) Runs a job with admin privileges

c) Starts a scheduled task

d) Starts a remote session

Answer: a) Starts a background job asynchronously

Explanation: Runs code in background, returning immediately.

37.

Which cmdlet allows querying for objects using XPath or XML?

a) Select-Xml

b) Get-Xml

c) Find-Xml

d) Query-Xml

Answer: a) Select-Xml

Explanation: Queries XML data using XPath expressions.

38.

How do you define a function named Get-Square that returns the square of a number?

powershell

Copy code

```
function Get-Square { param ($num) return $num * $num }
```

- a) Correct
- b) Incorrect, missing function keyword
- c) Incorrect, return not valid
- d) Incorrect syntax

Answer: a) Correct

Explanation: Proper function syntax with param and return.

39.

What is the use of Write-Output cmdlet?

- a) Sends objects down the pipeline
- b) Writes to console only
- c) Writes to log file
- d) Displays error messages

Answer: a) Sends objects down the pipeline

Explanation: Outputs to pipeline, can be captured or displayed.

40.

How do you declare an array with values 1, 2, 3 in PowerShell?

- a) \$arr = @(1,2,3)
- b) \$arr = [1,2,3]
- c) \$arr = {1,2,3}
- d) \$arr = (1;2;3)

Answer: a) \$arr = @(1,2,3)

Explanation: @() creates an array in PowerShell.